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Measuring Factual Knowledge in Text-to-Image Models

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Statement of authorship

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Abstract

Text-to-image (T2I) models produce convincing images from short prompts, but a convincing image is no evidence that the model knows the depicted entity. This thesis measures the factual knowledge of three T2I models, GPT Image 2, FLUX.2-dev, and FLUX.2-klein-4B, about 298 real-world entities: 100 curated entities from four domains, each annotated with an expected type and a distinctive visual feature, and 198 randomly sampled Wikipedia entities in three popularity tiers. Every image is generated from the prompt "An image of [entity]." A vision-language model, used as a judge, first describes each image without knowing which entity it should show and lists short statements about what is visible, called propositions; a second call checks these statements against the introduction of the entity's Wikipedia article. On the 289 entities available for all models, the models draw the correct coarse type for 73–79% of the curated entities, but the distinctive feature for only 63% (GPT Image 2), 46% (FLUX.2-dev), and 29% (FLUX.2-klein-4B). Of the extracted propositions, 9.0%, 9.7%, and 17.7% contradict the Wikipedia introduction; for GPT Image 2 this share rises from 3% for the most popular to 23% for the least popular entities. The most severe errors come from reading the entity name as something else, such as a portrait of a man for the British bus model Dennis Dart. Unlike language models, image generators cannot abstain, and their knowledge is limited to what can be drawn. Their outputs should therefore not be used as factual depictions without verification.

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1 Introduction

Text-to-image (T2I) models can produce detailed images from very short natural-language prompts. Their outputs may be visually coherent and aesthetically convincing, but visual plausibility does not imply factual knowledge. A model may generate a plausible castle, animal, or portrait without showing which properties distinguish the requested entity from related entities. It may also add contextual details that are not supported by the entity's real-world description.

This distinction matters when generated images are used for educational material, search interfaces, knowledge bases, or other multimodal applications. In these settings, an image may be interpreted as an assertion about the entity. An incorrect tower shape, an invented historical attribute, or an unrelated object can therefore turn a visually attractive image into misleading information.

1.1 Motivation

The problem becomes pressing when images are generated automatically for many entities at once. Brinster (2025) built such a pipeline: it illustrates the entities of a large knowledge base with images from generative vision models. A pipeline of this kind is only useful if the image generator knows the entities it is asked to draw. This thesis therefore examines this precondition and measures what image-generation models know about individual entities.

Measuring this is harder for an image generator than for a large language model (LLM). An LLM can be asked questions about an entity, and when it does not know the answer, it can say so. A text-to-image model has no such option: it always returns an image, and a model that does not know an entity still produces a polished picture of something, without signalling that it has guessed. What such a model knows can therefore only be read off the images it produces.

For the images to reflect the model's own knowledge, the prompt must not supply the facts that are being measured. All models in this thesis receive the same minimal prompt, "An image of [entity].", which contains only the entity name. Such a prompt is called *zero-context* here: it contains no Wikipedia text, no hints about the entity's appearance, and no other description. Every correct detail in the image must come from what the model learned during training.

(a) GPT Image 2

(b) FLUX.2-dev

(c) FLUX.2-klein-4B

Figure 1.1: Images generated from the prompt “An image of Dennis Dart.”

1.2 Problem Statement

The central question of this thesis is thus not whether a model can produce a convincing image, but whether the image is correct for the specific entity it was asked to show. The thesis examines this question for three current models: GPT Image 2 by OpenAI and two models of the FLUX.2 family by Black Forest Labs, the large FLUX.2-dev and the much smaller FLUX.2-klein-4B. The following example shows how far a convincing image and a correct one can diverge.

Figure 1.1 shows the three images generated for the entity *Dennis Dart*. Readers outside the United Kingdom will rarely know this name, which is what makes it a useful example. The Dennis Dart is a single-decker midibus chassis built by the British manufacturer Dennis from 1989 onwards; more than 12,600 were produced, most of them for British bus operators (Wikipedia contributors, 2026a). A correct image therefore shows a short single-deck bus, typically in the colours of a British operator. The name itself gives no hint of this: “Dennis” is a common first name and a “dart” is a small throwing object. A model that does not know the entity has to guess from these two words.

The three models guess differently. GPT Image 2 draws a single-deck bus with “DENNIS Dart” written on the front and the logo of the British operator First. FLUX.2-dev draws a red and grey hand-held device on a wooden floor, with “Dennis Dart” as a heading above a line of unreadable pseudo-text. FLUX.2-klein-4B draws a portrait of a smiling grey-haired man, reading “Dennis” as a person’s name. Both FLUX images are sharp and photorealistic; a reader who does not know what a Dennis Dart is cannot tell that they are wrong.

The example shows that an image can go wrong at different levels. It can show the wrong kind of object, as the device and the portrait do. It can show the right kind of object without the details that make it this particular entity, such as a generic city bus instead of a small single-deck bus. And it can contain details that are simply false. Because a convincing appearance hides all three kinds of error, the images have to be checked systematically. This thesis does so along the following research questions.

1.3 Research Questions

The thesis addresses three research questions:

RQ1: For how many real-world entities can T2I models correctly predict their type?

RQ2: For how many real-world entities do T2I models know descriptive features that distinguish them from related entities?

RQ3: Of the facts expressed in generated images for a given entity, how many are true?

All three questions are answered with the same procedure, which Chapter 3 describes in detail. Each model generates one image per entity from the zero-context prompt. A vision-language model (VLM), called the *judge* in this thesis, then describes each image without being told which entity it is supposed to show. Besides a short description and a guess of the object type, the judge writes down short, self-contained statements about what is visible, such as "The vehicle is a bus" or "The sky is clear and blue". These statements are called *propositions*; they make the content of an image countable. In a second step, the judge compares its output with the introduction of the entity's English Wikipedia article, called the *lead summary*. For RQ1 and RQ2, the comparison uses a type and a distinctive feature that were annotated in advance for a curated set of entities; for RQ3, the judge counts how many propositions the lead summary shows to be false.

1.4 Contributions

The thesis makes four contributions.

1. A benchmark of 298 real-world entities: 100 curated entities from four domains, each annotated with an expected type and a distinctive visual feature, and 198 randomly sampled Wikipedia entities in three popularity tiers defined by article size.
2. A public corpus of 885 images generated by GPT Image 2, FLUX.2-dev, and FLUX.2-klein-4B from the same zero-context prompt, together with a record of the nine prompts that OpenAI's safety system refused.
3. A two-step evaluation in which a vision-language judge first describes each image without knowing the entity and then checks the resulting propositions against the Wikipedia lead summary, yielding type match (RQ1), feature presence (RQ2), and the share of false propositions (RQ3).
4. An analysis of the results by domain and popularity, a catalogue of typical failure patterns, a comparison with published estimates of the factual knowledge of language models, and a critical assessment of the evaluation design with a proposal for a better one.

Code, images, evaluation results, and all derived tables and figures are available at `https://github.com/mrEkso/t2i-factuality-benchmark`.

1.5 Thesis Structure

Chapter 2 reviews text-to-image models, the evaluation of visual factuality, hallucinations, and what is known about factual knowledge in language models. Chapter 3 describes the benchmark, the generation setup, the two-step evaluation, and the metrics. Chapter 4 reports the results and shows examples of good and bad generations. Chapter 5 answers

the research questions, compares the findings with knowledge in language models, critically assesses the methodology, and discusses limitations. Chapter 6 concludes.

2 Related Work

This chapter places the thesis in the context of earlier work. It first describes how current text-to-image models work (Section 2.1) and how generated images have been used to illustrate knowledge bases (Section 2.2). It then reviews how the factual correctness of generated images has been evaluated (Section 2.3) and what is known about hallucinations and rare entities (Section 2.4). Section 2.5 summarizes how the factual knowledge of language models has been measured, and Section 2.6 compares this benchmark with the closest related work, in particular T2I-FactualBench.

2.1 Text-to-Image Models

Most current text-to-image models build an image step by step from random noise. Diffusion models learn to remove the noise gradually, guided by an encoding of the text prompt (Rombach et al., 2022; Podell et al., 2024). Rectified-flow models, a more recent variant, learn a straight path between noise and image (Esser et al., 2024). How closely the result follows the prompt depends on the text encoder and on the quality of the image captions used in training (Betker et al., 2023). The training objective rewards images that match their captions and look realistic; it does not reward factual correctness for a named entity. Whatever a model knows about an entity is therefore a by-product of its training data.

The three models studied in this thesis differ in how openly they are available. FLUX.2-dev and FLUX.2-klein-4B by Black Forest Labs are *open-weight* models: their trained parameters are public, so they can be run on one's own hardware. Both combine a language model, which encodes the prompt, with a rectified-flow transformer, which generates the image (Black Forest Labs, 2025, 2026). GPT Image 2 by OpenAI is a proprietary model. It runs only on OpenAI's servers and can be used only through OpenAI's programming interface (API); its architecture and size are not disclosed (OpenAI, 2026).

2.2 Knowledge Bases and Visual Representations

A knowledge base stores facts about entities, usually as subject–relation–object *triples* such as (Dennis Dart, manufacturer, Dennis). Many entities in public knowledge bases such as Wikidata have no image, and earlier work has used generative models to fill this gap:

Abu Ahmad et al. (2023) generated images for Wikidata entities from their triples, and Brinster (2025) illustrated a knowledge base that had itself been extracted from a language model. In both cases, the generated image serves as the illustration of the entity. This thesis asks the preceding question, namely whether such images can be trusted. It uses Wikipedia only as a source of entity names and reference descriptions and does not build or complete a knowledge base.

2.3 Evaluating Visual Factuality

Deciding whether a generated image can be trusted requires an evaluation of its content, not only of its quality. Image-quality metrics assess realism, and similarity scores based on CLIP, a model that maps images and texts into a shared representation, measure how well an image matches a caption (Radford et al., 2021; Hessel et al., 2021). Neither checks whether the image shows the correct entity or contains false attributes; surveys treat image quality, prompt alignment, aesthetics, and factual correctness as separate objectives (Hartwig et al., 2024). Broad benchmarks cover many such aspects at once: HEIM evaluates, among others, alignment, quality, and bias, and T2I-CompBench++ tests whether several objects and attributes named in one prompt are combined correctly (Lee et al., 2023; Huang et al., 2025a).

For prompt alignment, the field has moved from single similarity scores to question answering over atomic claims: TIFA and VQ^2 derive questions from the prompt and answer them with a visual question-answering (VQA) model, that is, a model that answers questions about an image (Hu et al., 2023; Yarom et al., 2023). The same idea has been applied to factuality. I-HallA measures factual errors in generated images with curated questions (Lim et al., 2025), T2I-FactualBench evaluates knowledge-intensive concepts with multi-round question answering (Huang et al., 2025c) (compared in detail in Section 2.6.1), KITTEN evaluates how faithfully generated images depict real-world entities (Huang et al., 2024), and FAGER builds a factual rubric from language-model knowledge and reference images (Lim et al., 2026). Large proprietary vision-language models such as GPT-4V have also been used directly as evaluators (Zhang et al., 2023). This thesis follows the claim-based approach with two differences. First, the claims are extracted from the image by a judge that does not know which entity the image should show. Second, they are checked against an external text reference, the Wikipedia lead summary. Instead of a proprietary model, the judge is Gemma 4 31B, an open-weight VLM by Google (Section 3.2), so that the evaluation can be re-run without a commercial API; Section 5.4.2 discusses the cost of this choice.

2.4 Hallucinations and Long-Tail Entities

When a generated image shows something false about the entity, this is called a *hallucination*, a term taken from text generation. Surveys of hallucination in text generation distinguish *intrinsic* hallucinations, which contradict the source, from *extrinsic* ones, which the source can neither confirm nor refute (Ji et al., 2023; Huang et al., 2025b). This thesis counts only the intrinsic kind: a proposition is a hallucination if the Wikipedia lead summary shows it to be false. A background that the summary does not mention, for example, is extrinsic and is not counted (Section 3.4).

Earlier work has also studied what makes such errors more likely and how to reduce them. Retrieving real reference images during generation has been proposed as a remedy (Lim

and Shim, 2024), and DoFaiR shows that instructions to increase demographic diversity can make images of historical subjects factually inaccurate (Wan et al., 2024). A further factor, and the one studied here, is how well known an entity is. For language models, this factor is well documented: their factual accuracy drops sharply for entities that are rarely mentioned on the web (Mallen et al., 2023; Kandpal et al., 2023). Image generators may behave in the same way, since they have seen few or no pictures of rare entities during training. To test this, the benchmark includes a *Long-tail* tier of entities with very short Wikipedia articles, which serve as a proxy for low popularity (Section 3.1).

2.5 Factual Knowledge in Language Models and Image Generators

For language models, the amount of factual knowledge has been estimated directly. GPTKB materializes the knowledge of GPT-4o-mini by recursively prompting the model for facts about entities; the resulting knowledge base contains 101 million triples about 2.9 million entities (Hu et al., 2025b). The authors relate this number to model size: at an estimated 8 billion parameters, GPT-4o-mini uses about 79 parameters per retrieved triple. A later version built from GPT-4.1 contains 100 million triples about 6.1 million entities (Hu et al., 2025a). Controlled experiments on synthetic data give an upper bound: a language model can store about two bits of knowledge per parameter (Allen-Zhu and Li, 2025).

No comparable estimate appears to exist for image generators, and their knowledge differs from that of LLMs in three ways. First, the unit is different. An LLM states a fact as a sentence that can be mapped to a subject–relation–object triple. An image states facts implicitly, and only facts that are visible: a platypus's bill or a bus livery can be drawn, a birth date or an election result cannot, except as written text. The facts counted in this thesis are therefore visual propositions (Section 3.4), not triples. Second, the elicitation is different. GPTKB asks many questions per entity and follows links to further entities; a zero-context image prompt gives the model one attempt per entity, and the model cannot abstain. Third, the knowledge is stored in different components. As described in Section 2.1, the FLUX models contain a language model: FLUX.2-dev couples a 24-billion-parameter Mistral vision-language model, which Black Forest Labs describe as the source of "real world knowledge", with a 32-billion-parameter rectified-flow transformer that produces the image (Black Forest Labs, 2025). FLUX.2-klein-4B pairs a 4-billion-parameter transformer with the Qwen3-4B language model (Black Forest Labs, 2026). OpenAI does not disclose the size or architecture of GPT Image 2 (OpenAI, 2026); its predecessor was introduced as image generation built into GPT-4o, which lets the model "link its knowledge between text and images" (OpenAI, 2025). An image generator can therefore fail in two places: its language component may not know the entity, or it may know the entity while the image component cannot draw what it knows.

2.6 Positioning of This Work

Table 2.1 compares this benchmark with the most closely related work. The numbers are taken from the respective papers and count different units: entities in this work and KITTEN, concepts in T2I-FactualBench, scenarios in HEIM, and test prompts in DoFaiR.

This benchmark is smaller than the broad evaluation suites. It differs from them in three respects: the prompt contains only the entity name, the same two-step evaluation with a

Table 2.1: Comparison with related knowledge-intensive and factuality benchmarks.

Benchmark	Test items	Models	Evaluation	Release/access
This work	298 entities	3	Blind VLM description, Wikipedia-grounded proposition checks	Public code, images, results
T2I-FactualBench	1,600 concepts (3,000 prompts)	7 (+4 variants)	Multi-round VQA with GPT-4o and a reference image; human validation	Benchmark and code
KITTEN	322 entities	7	Knowledge-intensive entity evaluation with automatic and human metrics	Benchmark/paper release
HEIM	62 scenarios	26	Holistic alignment, quality, safety, and bias evaluation	Benchmark suite
DoFaiR	756 test prompts	DALL-E 3	Factuality effects of diversity interventions	Benchmark and code

blind first call is applied to all models, and the release contains the full image corpus together with a record of the images that could not be generated. In addition, two thirds of its entities are sampled at random from Wikipedia and stratified by popularity instead of being chosen by hand.

2.6.1 Comparison with T2I-FactualBench

T2I-FactualBench (Huang et al., 2025c) is the work closest to this thesis. Its first level, *single knowledge concept memorization*, uses the same prompt template as this thesis, "An image of [concept]", and asks the same question: does a model draw a concept correctly when the prompt gives only its name? Two further levels go beyond this thesis. They place a concept in a given action, attribute, or scene, or combine two or three concepts in one image, for 3,000 prompts in total. Table 2.2 compares the two benchmarks point by point.

The benchmarks differ first in what they test. T2I-FactualBench defines a *knowledge concept* as a concept with fewer than four hyponyms in the BabelNet knowledge base, such as a dog breed, a plant species, a handbag model, a landmark, or a celebrity, and keeps eight categories that can be drawn. Many of its concepts are classes rather than individual entities, and the authors note that popularity influences the results without measuring this influence. This thesis uses only individual Wikipedia entities. Two thirds of them are drawn at random, which brings in abstract topics such as *Marxism* or a season of a football club and makes popularity an explicit variable (Section 3.1).

The second difference is the reference. T2I-FactualBench gives GPT-4o the generated image together with a reference image of the concept and asks for four binary scores. A reference image can verify appearance that no text describes, such as a face or the pattern of a fur; the Wikipedia lead summary used here usually cannot (Section 5.4.2). A reference image cannot, however, verify relations to other entities, such as the national team on a cricketer's shirt (Table 3.3). In addition, the texture score partly measures image quality, since it asks for textures "free from blurriness, pixelation, or artificial effects". A text reference covers relations, but not appearance, so the two references complement each other.

Third, the evaluator of T2I-FactualBench knows which concept the image should show. This thesis keeps the first judge call blind so that the entity name cannot suggest what the judge should see. The price is that a feature that is drawn but not described in the blind call is lost (Section 3.3).

Table 2.2: T2I-FactualBench (Huang et al., 2025c) compared with this benchmark. The T2I-FactualBench column refers mainly to its memorization level, which corresponds to the task studied here.

	T2I-FactualBench	This work
Test items	1,600 concepts in eight categories (animals, plants, persons, locations, artifacts, food, events, celestial bodies)	298 individual entities: 100 curated in four domains, 198 sampled from Wikipedia
Selection	Concepts with fewer than four hyponyms in BabelNet, many of them classes such as breeds or product types; abstract categories excluded	Curated entities chosen by hand; Random entities drawn at random, including abstract topics
Popularity	Acknowledged as a factor, not measured	Three tiers by article size
Prompts	"An image of [concept]", plus instantiation and composition prompts	"An image of [entity]" only
Models	SD v1.5, SDXL, SD 3.5, PixArt-α, Playground v2.5, FLUX.1-dev, DALL-E 3, and four knowledge-injection variants	GPT Image 2, FLUX.2-dev, FLUX.2-klein-4B
Evaluator	GPT-4o; sees the concept name, the generated image, and a reference image	Gemma 4 31B; describes the image blind, then checks the description without the image
Reference	One image per concept from BabelNet	Wikipedia lead summary
Score	Four binary scores (shape, colour, texture, feature details), averaged	Type match, feature presence, share of contradicted propositions
Human validation	Three annotators, 900 samples (300 for concept factuality)	None

Fourth, T2I-FactualBench validates its evaluator. Three annotators rated 900 samples, 300 of them for concept factuality, and on these the concept scores of GPT-4o reach a Spearman correlation of 0.57 with the mean human rating. Without the reference image the correlation drops to 0.44, and the best open-weight evaluator, Qwen2.5-VL-72B, reaches 0.49. The judge of this thesis has not been validated in this way, and these numbers suggest that an open-weight judge without a reference image is the weaker setup (Section 5.3).

Where the results can be compared, they agree in direction. At the memorization level, the proprietary DALL-E 3 scores highest (55.5 of 100, without persons), ahead of Stable Diffusion 3.5 (46.2) and FLUX.1-dev (35.6), just as GPT Image 2 leads on RQ2 here. The authors also report the same kind of near miss as in Table 4.6, for example a wild boar instead of a Baird's tapir, and DALL-E 3 could not be evaluated on persons because of OpenAI's content policy, which matches the refusals of GPT Image 2 (Section 3.5). One result differs: T2I-FactualBench finds locations among the hardest categories, whereas landmarks have the highest feature rates in this thesis. Its location category, however, also contains natural landforms, while the curated landmarks here are world-famous buildings.

The two benchmarks therefore complement each other. T2I-FactualBench covers more concepts and harder prompts and validates its evaluator against human ratings. This thesis covers individual entities sampled by popularity, keeps the first evaluation step blind, and checks the images against an external text reference.

3 Methodology

This chapter describes how the benchmark was built and evaluated. Section 3.1 introduces the entities, Section 3.2 the image generators, the judge, and the generation setup, and Section 3.3 the two-step evaluation. Section 3.4 explains which kinds of propositions the judge produces, and Section 3.5 how missing images are handled. The last two sections describe two additional analyses: the relation between the number of propositions and the error rate (Section 3.6) and an extrapolation of the results to the whole English Wikipedia (Section 3.7).

3.1 Benchmark Design

The benchmark consists of 298 entities in two subsets that serve different purposes (Table 3.1). The *curated* subset contains 100 entities selected by hand, for which the expected appearance is annotated; it is used for all three research questions. The *Random* subset contains 198 entities sampled from Wikipedia; it has no annotations and is used for RQ3 and for the analysis by popularity.

Table 3.1: Composition of the factuality benchmark.

Subset or category	Entities	RQ1/RQ2 annotations
Famous Landmarks and Architecture	25	Type and feature
Historical Figures	25	Type and feature
Rare Flora and Fauna	25	Type and feature
Specific Cultural Artifacts and Foods	25	Type and feature
Random: Famous tier	66	Not defined
Random: Medium tier	66	Not defined
Random: Long-tail tier	66	Not defined
Total	298	–

The curated entities come from four domains with 25 entities each: famous landmarks and architecture, historical figures, rare flora and fauna, and specific cultural artifacts and foods. The domains cover buildings, people, living beings, and objects, so that knowledge concentrated in one domain can be separated from general visual competence. Within each domain, the author chose entities that are recognizable but differ in how familiar they

are. Each curated entity is annotated with an expected type and a distinctive feature. The expected type names the class of the entity (for example, “Castle” for Neuschwanstein or “Mammal” for Platypus). The distinctive feature is a visible property that separates the entity from similar ones (for example, “duck-like bill and beaver-like tail” for Platypus). The complete list is given in Appendix A.4.

The Random subset tests entities that were not chosen by hand. Its entities were drawn with the MediaWiki `list=random` API from the article namespace of the English Wikipedia; titles containing “List of” or “disambiguation” were skipped. Each drawn article was assigned to a popularity tier by its size, measured in bytes of wikitext (the article’s source text) as returned by the MediaWiki API and released as the `Wiki_Length` field: Famous for at least 100,000 bytes, Medium for 20,000 to less than 100,000 bytes, and Long-tail for 500 to less than 20,000 bytes. Sampling continued until each tier held 66 entities, so that no tier dominates the sample.

Article size serves as a proxy for popularity: widely known topics attract more editors and references, and their articles grow accordingly. The proxy is imperfect, since a short article can also describe an important but little-documented topic. The tiers are therefore strata of the sample, not a claim about how often an entity occurs on the web.

3.2 Models and Image Generation

The three image generators cover the two settings in which such models are used in practice: a proprietary service and openly available models. GPT Image 2 is the image model that OpenAI released in April 2026; it runs only on OpenAI’s servers and filters requests with its own safety system. FLUX.2-dev and FLUX.2-klein-4B are open-weight models from the same family, FLUX.2-klein-4B being the much smaller one. Comparing the two FLUX models shows the effect of model size within one family, and comparing them with GPT Image 2 shows how open models relate to a commercial one. The selection is not meant as a general leaderboard.

Table 3.2 lists the published sizes of the three generators and, for comparison, of the language model analysed in GPTKB (Section 2.5).

Table 3.2: Size of the evaluated image generators and of the LLM used for GPTKB. Parameter counts are those published by the providers; the GPT-4o-mini value is the estimate used by the GPTKB authors.

Model	Language component	Image component	Access
GPT Image 2	not disclosed	not disclosed	OpenAI API only
FLUX.2-dev	Mistral VLM, 24B	rectified-flow transformer, 32B	open weights
FLUX.2-klein-4B	Qwen3-4B, 4B	rectified-flow transformer, 4B	open weights
GPT-4o-mini (GPTKB)	about 8B (estimate)	–	OpenAI API only

As described in Section 1.1, every entity is rendered with the same zero-context prompt:

```
An image of [entity].
```

GPT Image 2 was called through OpenRouter, a service that forwards requests to the OpenAI API (model `openai/gpt-image-2`), at 1024×1024 pixels. The two FLUX models are run by the Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI) Dresden/Leipzig and were called through its image-generation API with the default settings; they also return

1024×1024 images. Each model generated one image per entity, without a fixed random seed. The generation script skips existing files, so interrupted runs can be resumed without repeating requests.

The judge used in the evaluation (Section 3.3) is Gemma 4 31B in its instruction-tuned version (`gemma-4-31B-it`), an open-weight VLM by Google with about 31 billion parameters that accepts both text and images (Google DeepMind, 2026). It is also served by ScaDS.AI. An open-weight judge was chosen so that the complete evaluation can be re-run from the public release without a paid commercial API.

3.3 Visual Evaluation Pipeline

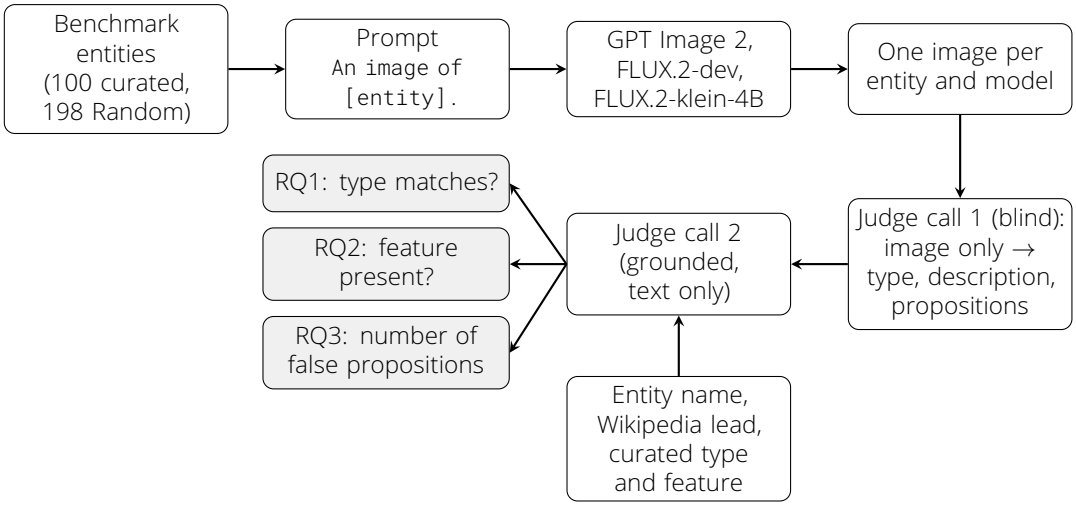

Figure 3.1: Benchmark pipeline: generation, blind judge call, and grounded judge call.

Each image is evaluated in two calls to the judge (Figure 3.1). Both calls use temperature 0, which makes the output as deterministic as possible, and return machine-readable JSON; Appendix A.3 gives the exact prompts. In the first, *blind* call, the judge sees only the image, not the entity name, and returns a predicted type, a short description, and a list of propositions about what is visible. Keeping this call blind prevents the entity name from suggesting what the judge should see. The second, *grounded* call does not see the image. It receives the entity name, the Wikipedia lead summary, the curated type and feature (for curated entities), and the output of the blind call. It returns the decisions for RQ1 and RQ2 and the number of propositions that the lead summary shows to be false. Wikipedia titles are resolved through the MediaWiki search API, and the retrieved lead summaries are cached in the released evaluation directory. All 885 successful evaluations use a Wikipedia lead summary as reference; for two curated entities the search returned the wrong article (Section 5.4.2).

For RQ1, a result is correct if the grounded call accepts the type predicted in the blind call as compatible with the curated expected type. The blind prompt offers broad example labels (Building, Person, Animal, Landscape, Object, None) but does not restrict the answer to them, while the curated types are more specific. A broad label therefore counts as a match when it covers the expected type: "Building" matches "Castle" for Neuschwanstein, and "Animal" matches "Mammal" for Platypus. A label from a different class, such as "Person" for a pastry, is a mismatch. RQ2 is decided separately: it is correct only if the blind description or propositions clearly indicate the annotated distinctive feature. Since the grounded call sees only this text, a feature that is drawn but not mentioned in the blind call counts as absent.

Keeping RQ1 and RQ2 apart prevents a broad type match from being taken as recognition of the entity.

For RQ3, the decision is made per proposition. A proposition that the lead summary shows to be false, or that contradicts it, is counted as a hallucination (Section 2.4); a proposition that the summary does not mention is not counted. Counting per proposition avoids treating an image with ten correct claims and one false claim like an image with a single false claim.

The counts are aggregated per model into the main metric, the hallucination rate

$$\mathrm{HallucinationRate}(m) = \frac{\sum_i H_{i,m}}{\sum_i P_{i,m}}, \tag{3.1}$$

where $H_{i,m}$ is the number of hallucinations in image i generated by model m, and $P_{i,m}$ is the number of propositions extracted from that image. This is a pooled rate: it sums hallucinations and propositions before dividing, so an image with a very short description does not receive the same weight as a longer one. The Spearman analysis in Section 4.4 uses the per-image rate $H_{i,m}/P_{i,m}$ instead.

All rates are reported with 95% Wilson confidence intervals, a standard interval for proportions that remains reliable for small samples and for rates close to 0 or 1. Differences between two models in RQ1 and RQ2 are tested with two-sided two-proportion z-tests.

3.4 What Counts as a Visual Fact

A language model states a fact in a sentence; an image states facts only implicitly. The blind call makes them explicit as propositions (Section 1.3), which are the unit counted in RQ3. Not all propositions carry the same kind of information: those in the corpus are of five kinds. Table 3.3 gives one example of each, taken from the evaluation file.

Table 3.3: Kinds of propositions extracted from generated images, with examples from the evaluation corpus.

Kind	Example proposition (entity, model)	Checkable against the lead summary?
Type	"The vehicle is a bus" (Dennis Dart, GPT Image 2)	Yes, almost always
Appearance	"There are multiple vertical fluted columns" (Parthenon, FLUX.2-dev)	Often, if the summary describes the appearance
World knowledge	"There is a silver fern logo and the text 'NEW ZEALAND' on the left side of the chest" (Heath Streak, GPT Image 2)	Sometimes; here it contradicts "Zimbabwean cricketer"
Rendered text	"The text 'DENNIS Dart' is written on the front" (Dennis Dart, GPT Image 2)	Rarely; it mostly repeats the prompt
Scene	"The sky is clear and blue" (Parthenon, FLUX.2-dev)	Almost never

Type and appearance propositions carry most of the entity knowledge. World-knowledge propositions are the most informative ones: a jersey, a flag, or a bus livery links the entity to another entity (a country, an operator, a place), which is the kind of fact a knowledge base stores as a triple. Rendered text and scene propositions carry little knowledge. A keyword rule applied to all 6,439 propositions of the common set (the 289 entities generated by all three models; Section 3.5) gives a rough picture: 18–24% of the propositions describe the background, lighting, or weather, and 9% (FLUX models) to 20% (GPT Image 2) transcribe

text written in the image. The remaining 61–68% describe the depicted object itself. The rule is simple, so these shares are approximate.

The distinction matters for RQ3. Since only propositions shown to be false are counted, scene and text propositions are almost never counted as errors, and an image that shows only a generic landscape can reach a hallucination rate of zero (Section 4.6).

3.5 Common Set and Missing Evaluations

Not every model produced an image for every entity: GPT Image 2 refused nine of the 298 prompts through its safety system. Comparing each model on all of its own images would therefore compare the models on different entities. The main comparison instead uses the *common set*, the 289 entities for which all three models produced an image. The nine refusals are recorded separately and counted neither as correct nor as incorrect, so that a policy decision of the provider is not mistaken for missing knowledge. Results on all images of each model, called *available-case* results, are reported only where they show that the restriction to the common set does not change a conclusion.

3.6 Proposition-Count Analysis

The hallucination rate depends on how many propositions an image yields: a short description offers fewer chances to be wrong, but also says less about the entity. The analysis therefore reports the average number of propositions per image by model and popularity tier alongside the rates. Spearman rank correlation is used to check whether images with more propositions also have a larger share of false ones. A coefficient near zero indicates no consistent monotonic relationship, and values close to 1 or -1 indicate a strong one.

3.7 Weighted Extrapolation to the Full Encyclopedia

The last analysis estimates how many non-contradicted visual propositions a model would produce if it drew every entity of the English Wikipedia. This puts the results on the same scale as the estimates of LLM knowledge discussed in Section 2.5. The estimate cannot be obtained by averaging the three tiers with equal weights, because the benchmark samples the tiers equally by design, while most Wikipedia articles are short. Such a naive average would overweight the popular entities.

To correct for this, the population share w_t of each tier was estimated from Wikipedia's published statistics on article size (Wikipedia contributors, 2026b). These statistics are expressed in words rather than bytes of wikitext, so the benchmark thresholds were converted with a rough factor of 7 bytes per word; the conversion is used only for this estimate, not for assigning entities to tiers. The resulting shares are about 0.32% (Famous), 5.3% (Medium), and 94.3% (Long-tail). This agrees with the fact that about 32% of all English Wikipedia articles are stubs, that is, articles too short to cover their topic adequately (Wikipedia contributors, 2026d). Because wikitext markup varies across articles, w_t is an order-of-magnitude estimate rather than an exact population statistic.

The weighted extrapolation for model m is

$$\mathrm{Total}(m) = N_{\text{total}} \times \sum_{t \in \{F,M,L\}} w_t \cdot \bar{F}_{t,m} \cdot \left(1 - \mathrm{HallucinationRate}_{t,m}\right), \tag{3.2}$$

where $N_{\text{total}} = 7{,}237{,}727$ is the number of English Wikipedia articles on 18 September 2026 (Wikipedia contributors, 2026c), $\bar{F}_{t,m}$ is the average number of propositions per image for tier t and model m, and $\mathrm{HallucinationRate}_{t,m}$ is the pooled hallucination rate for the same tier and model; both are reported in Chapter 4. The product of the last two factors is the number of propositions per image that are not shown to be false. It includes scene and text propositions (Section 3.4), so $\mathrm{Total}(m)$ is an upper bound on the number of correct entity facts, not a count of verified facts. Section 5.3 discusses how well this quantity reflects what a model knows.

4 Results

This chapter reports the results in the order of the research questions. Section 4.1 gives the coverage of the three models and an overview of all measures. Section 4.2 reports type and feature recognition (RQ1 and RQ2), and Sections 4.3 and 4.4 report the hallucination rates and the number of propositions per image (RQ3). Section 4.5 extrapolates the results to the English Wikipedia, and Section 4.6 shows examples of good and bad generations.

4.1 Coverage and Overall Comparison

The evaluation file contains 885 successful image evaluations: 298 for each FLUX model and 289 for GPT Image 2. The nine missing GPT Image 2 images were refused by OpenAI's safety system (Table A.1). Three of them are curated historical figures (Mahatma Gandhi, Frida Kahlo, and Salvador Dalí); the other six are Random entities, four of them films or television episodes. As described in Section 3.5, the refusals are not scored as errors, and all comparisons below use the common set of the 289 entities generated by every model: 97 curated entities and 192 Random entities (64 Famous, 63 Medium, and 65 Long-tail). Table 4.1 summarizes all three research questions on this set.

Table 4.1: Overall comparison on the common set. RQ1 and RQ2 are computed on the 97 curated entities, the other columns on all 289 entities. Values in parentheses are 95% Wilson confidence intervals.

Model	RQ1	RQ2	Avg. hallucinations	Avg. propositions	RQ3 pooled rate
GPT Image 2	77.3% (68.0–84.5)	62.9% (53.0–71.8)	0.70	7.70	9.0% (8.0–10.3)
FLUX.2-dev	79.4% (70.3–86.2)	46.4% (36.8–56.3)	0.69	7.13	9.7% (8.4–11.0)
FLUX.2-klein-4B	73.2% (63.6–81.0)	28.9% (20.8–38.6)	1.32	7.45	17.7% (16.1–19.4)

No model is best on every measure. FLUX.2-dev has the highest type-match rate (RQ1), GPT Image 2 the highest feature rate (RQ2), and GPT Image 2 and FLUX.2-dev have similar hallucination rates (RQ3), while the rate of FLUX.2-klein-4B is nearly twice as high. The RQ3 columns cover all 289 entities, curated and Random, and therefore differ from the tier values in Table 4.2, which cover only the 192 Random entities.

4.2 Type and Feature Recognition

RQ1 and RQ2 are measured on the 97 curated entities of the common set, called the *common curated subset* below. On this subset, the type-match rates are 77.3% for GPT Image 2, 79.4% for FLUX.2-dev, and 73.2% for FLUX.2-klein-4B. The available-case rates are almost the same (77.3%, 80.0%, and 73.0%), so the restriction to the common set does not change the picture.

FLUX.2-dev thus has the highest type-match rate, but the differences are small: none of the three pairwise tests is significant for RQ1 (GPT Image 2 versus FLUX.2-dev $p = 0.727$, GPT Image 2 versus FLUX.2-klein-4B $p = 0.506$, and FLUX.2-dev versus FLUX.2-klein-4B $p = 0.311$). The feature rates differ much more. GPT Image 2 shows the distinctive feature for 62.9% of the entities, FLUX.2-dev for 46.4%, and FLUX.2-klein-4B for 28.9%. All three pairwise differences are significant without a correction for multiple testing (GPT Image 2 versus FLUX.2-dev $p = 0.021$, GPT Image 2 versus FLUX.2-klein-4B $p < 0.001$, and FLUX.2-dev versus FLUX.2-klein-4B $p = 0.012$). With a Bonferroni correction for three comparisons, which lowers the threshold to $0.05/3 \approx 0.0167$, the difference between GPT Image 2 and FLUX.2-dev is no longer significant, while the other two remain so. Recognizing the broad category is therefore not the same as showing the feature that makes an entity visually distinctive. Figures 4.1 and 4.2 break both rates down by curated category.

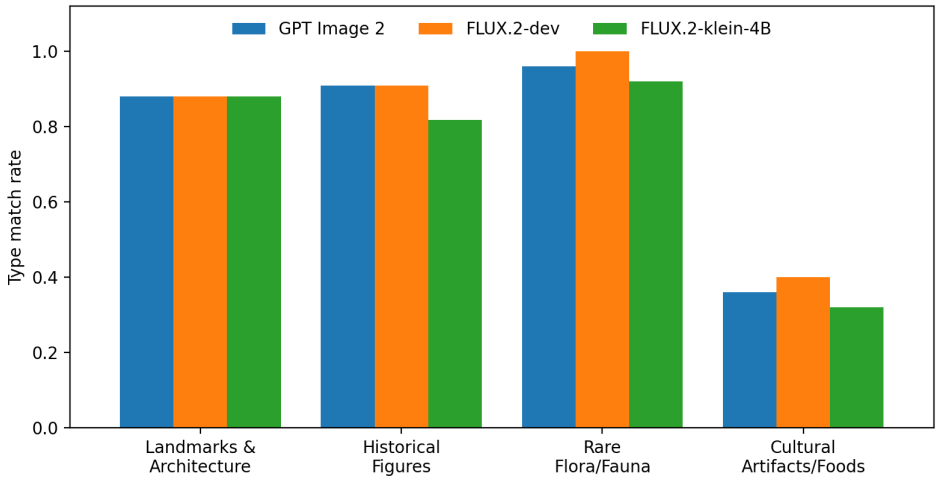

Figure 4.1: Type-match rate (RQ1) by curated category on the common curated subset.

The figures show why the overall means should be read together with the category results. Type recognition is highest for flora and fauna (92–100%) and lowest for cultural artifacts and foods (36% for GPT Image 2, 40% for FLUX.2-dev, and 32% for FLUX.2-klein-4B). The low value for artifacts and foods says more about the labels than about the images. In the blind call, the judge labels 19 to 23 of the 25 artifact and food images per model as “Object”, and the grounded call accepts this label for an expected type such as “Food”, “Pastry”, or “Sword” in only 27 of 64 cases. The feature rates for the same category are much higher (76%, 60%, and 32%), so the models often do draw the distinctive feature of the artifact or dish.

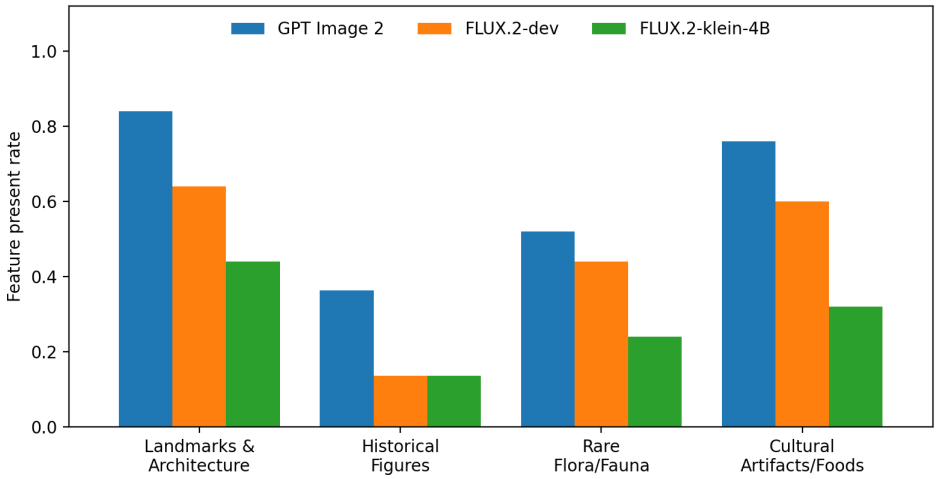

Figure 4.2: Distinctive-feature rate (RQ2) by curated category on the common curated subset.

4.3 Hallucination Rates

On the common set, 9.0% of the propositions extracted from GPT Image 2 images contradict the lead summary, 9.7% for FLUX.2-dev, and 17.7% for FLUX.2-klein-4B (Table 4.1). Per image, this corresponds to 0.70, 0.69, and 1.32 false propositions on average. Table 4.2 splits the rate by popularity tier.

Table 4.2: Pooled hallucination rate by popularity tier for the 192 Random entities of the common set. Values in parentheses are 95% Wilson confidence intervals.

Popularity tier	GPT Image 2	FLUX.2-dev	FLUX.2-klein-4B
Famous	3.0% (1.9–4.8)	10.9% (8.4–14.0)	22.8% (19.3–26.7)
Medium	7.3% (5.4–9.9)	8.0% (5.9–10.9)	15.9% (12.9–19.4)
Long-tail	23.0% (19.5–26.9)	14.6% (11.6–18.2)	21.1% (17.7–24.9)

Popularity affects the models differently. For GPT Image 2 the rate rises from 3.0% for Famous to 7.3% for Medium and 23.0% for Long-tail entities, and the confidence intervals of the three tiers do not overlap. FLUX.2-dev is lowest in the Medium tier (8.0%) and highest in the Long-tail tier (14.6%). FLUX.2-klein-4B is error-prone in every tier and, unexpectedly, highest for Famous entities (22.8%): nine of its 64 Famous images have every proposition judged false, and five of them show a single person for an abstract topic such as *Marxism* or the *112th United States Congress*. Only GPT Image 2 shows the clear popularity effect that the Long-tail tier was designed to test.

4.4 Number of Propositions per Image

All three models produce slightly fewer propositions for less popular entities (Figure 4.3). GPT Image 2 produces 8.33, 8.25, and 7.69 propositions per image for Famous, Medium, and Long-tail entities, FLUX.2-dev 7.61, 7.30, and 6.95, and FLUX.2-klein-4B 7.75, 7.70, and 7.51. The decrease from Famous to Long-tail is about 0.6 propositions for GPT Image 2 and FLUX.2-dev and 0.2 for FLUX.2-klein-4B. The higher Long-tail hallucination rate of GPT

Image 2 therefore does not come from longer descriptions: its Long-tail images contain fewer propositions, but a larger share of them is false.

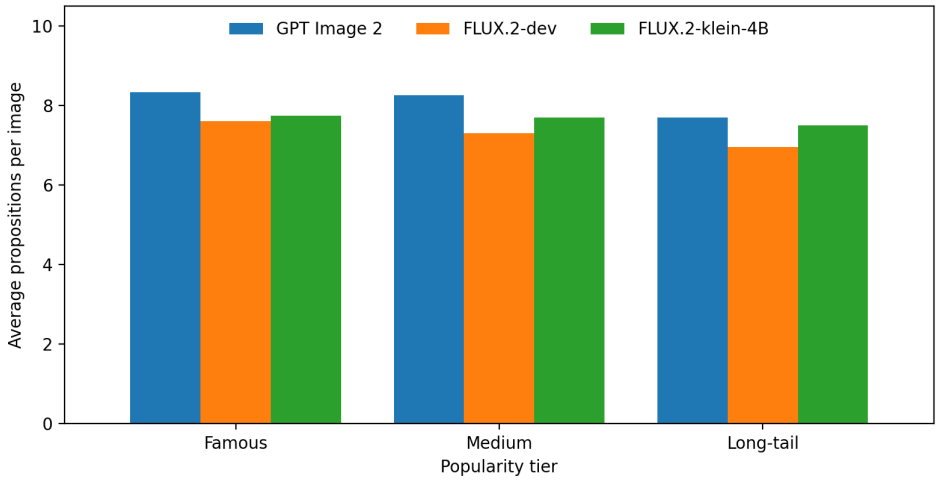

Figure 4.3: Average number of propositions per image by popularity tier (192 Random entities of the common set).

The relationship between the number of propositions and the hallucination rate is weak: the Spearman correlations between the two, computed per image, are 0.10 for GPT Image 2, 0.06 for FLUX.2-dev, and 0.08 for FLUX.2-klein-4B. Images with longer descriptions are thus not systematically less accurate, and shorter descriptions are not systematically more accurate. For this reason, the number of propositions and the hallucination rate are reported together.

4.5 Weighted Extrapolation Results

Table 4.3 applies Equation 3.2 to the common set. The naive estimate, which weights the three tiers equally, exceeds the population-weighted estimate by 20.0% for GPT Image 2, 8.4% for FLUX.2-dev, and 3.0% for FLUX.2-klein-4B. The difference is largest for GPT Image 2 because its error rate rises most steeply towards the Long-tail tier, which makes up about 94% of the weighted population. After weighting, the three models are almost equal, at 43.5, 43.3, and 43.0 million non-contradicted propositions. A stratified sample like this one therefore has to be reweighted before it is extrapolated, and the size of the correction depends on the model.

The near equality of the weighted totals stands in contrast to the clear differences between the models in RQ2 and in the hallucination rates. It arises in the Long-tail tier, which contributes 93–94% of every model's total. There, the three models produce almost exactly the same number of non-contradicted propositions per image (5.92 for GPT Image 2, 5.94 for FLUX.2-dev, and 5.92 for FLUX.2-klein-4B): GPT Image 2 has the highest error rate in this tier, but also the longest descriptions. In the Famous tier, GPT Image 2 is clearly ahead (8.08, 6.78, and 5.98 non-contradicted propositions per image), but this tier carries only 0.3% of the weight. The equal totals are therefore not evidence that the three models know equally much; Section 5.3 examines what the measure captures and how a better design would look.

Table 4.3: Naive versus population-weighted extrapolation of non-contradicted propositions over English Wikipedia ($N_{\text{total}} = 7{,}237{,}727$).

Model	Naive (equal-weight)	Population-weighted	Relative overstatement
GPT Image 2	52.2M	43.5M	+20.0%
FLUX.2-dev	46.9M	43.3M	+8.4%
FLUX.2-klein-4B	44.4M	43.0M	+3.0%

4.6 Examples of Good and Bad Generations

The aggregate numbers do not show what a good or a bad generation looks like, nor what the judge actually writes and decides. This section therefore gives the complete judge output for three entities of very different kinds: an animal (Platypus), a person (Boudica), and a city (Belgrade). It then summarizes the recurring patterns of bad generations. The examples illustrate how the metrics come about and are not used as additional statistical evidence.

4.6.1 An Animal: Platypus

Figure 4.4 shows the three images for *Platypus*, a curated entity with the expected type "Mammal" and the distinctive feature "duck-like bill and beaver-like tail". The three images separate the three research questions. GPT Image 2 draws a brown semi-aquatic mammal with a duck-like bill and webbed feet: type, feature, and all six propositions are correct. FLUX.2-dev draws a brown semi-aquatic mammal with a flat tail but a rounded dark snout instead of a duck-like bill. The type is correct and none of its seven propositions contradicts the summary, but the feature that separates a platypus from an otter or a beaver is missing, so RQ2 fails. FLUX.2-klein-4B draws a capybara: the type "Animal" is still accepted, but the feature is missing and all five propositions are judged false. In the sense of this benchmark, a generation is good only if it passes all three checks, as the GPT Image 2 platypus does.

(a) GPT Image 2

(b) FLUX.2-dev

(c) FLUX.2-klein-4B

Figure 4.4: Images generated from the prompt "An image of Platypus."

Table 4.4 shows what the judge produces for these images. The blind call returns five to seven propositions per image, most of them about the animal and one or two about its surroundings. The grounded call returns the two decisions, the number of false propositions, and a short explanation; it does not label each proposition separately. Two details of the table matter for the metrics. First, the RQ2 decision depends on the wording of the blind call: "a wide, flat snout resembling a bill" (GPT Image 2) counts as the duck-like bill, "a flat, wide

Table 4.4: Complete judge output for the Platypus images in Figure 4.4. The left column gives the type predicted in the blind call and the decisions of the grounded call; "false" is the number of propositions judged false out of all propositions. The right column lists the propositions of the blind call verbatim and the explanation returned by the grounded call. Propositions marked ✗ are those that the explanation names as false.

Image and decisions	Propositions (blind call) and explanation (grounded call)
GPT Image 2 type: Animal RQ1 ✓ RQ2 ✓ false: 0/6	(1) The animal has brown fur. (2) The animal has a wide, flat snout resembling a bill. (3) The animal has webbed feet. (4) The animal has a broad, flat tail. (5) The animal is positioned in shallow water with a pebbled bottom. (6) There is a brown leaf on the ground to the right of the animal. *Explanation:* none.
FLUX.2-dev type: Animal RQ1 ✓ RQ2 ✗ false: 0/7	(1) The animal has brown fur. (2) The fur appears wet. (3) The animal is in shallow water. (4) The animal has a flat, wide snout. (5) A small part of the tongue is visible. (6) The animal has small ears. (7) There are ripples in the water around the animal. *Explanation:* none.
FLUX.2-klein-4B type: Animal RQ1 ✓ RQ2 ✗ false: 5/5	(1) The animal has brown, coarse fur. ✗ (2) The animal has a large, dark snout. ✗ (3) The animal is sitting on green grass and dirt. ✗ (4) The animal has small ears. ✗ (5) The animal has long whiskers. ✗ *Explanation:* "The image depicts a capybara; the ground truth describes a platypus with a duck-bill and electrolocation, not a large dark snout, long whiskers, or the specific morphology of a capybara."

snout" (FLUX.2-dev) does not. Second, for the capybara the judge counts all five propositions as false, including "The animal has small ears" and "The animal is sitting on green grass and dirt", which the summary does not contradict. Once the judge concludes that the image shows a different animal, it rejects every proposition. For a wrong referent this is reasonable, but it means that the count is not a strict check of each proposition against the reference.

4.6.2 A Person: Boudica

Boudica is a curated historical figure with the expected type "Queen" and the distinctive feature "long red hair and ancient tribal clothing". Her lead summary describes what she did, not how she looked: she was a queen of the British Iceni tribe who led a failed uprising against the Roman Empire in AD 60 or 61. Figure 4.5 shows the images and Table 4.5 the judge output.

GPT Image 2 draws a red-haired woman with blue face paint and a plaid cloak in a carved chariot, in front of a banner that reads "BOUDICA"; all three checks pass. FLUX.2-dev draws a woman in classical, Roman-looking armour with a plumed helmet and a round shield, in front of an army of similarly armoured soldiers. The type is correct and the feature is missing. The judge counts four of nine propositions as false because Roman-style armour contradicts her role as the leader of an uprising against Rome. This is an inference from the summary, not a literal contradiction, and since the explanation names only the helmet and the breastplate with greaves, the other two propositions in the count cannot be identified. FLUX.2-klein-4B draws a bearded man in furs with a crown and two spears and writes the garbled text "BUULDCTH" into the image. The blind call predicts "Person", but the grounded call rejects the type, because a bearded man is not a queen, and counts the two propositions about the man and his beard as false.

(a) GPT Image 2

(b) FLUX.2-dev

(c) FLUX.2-klein-4B

Figure 4.5: Images generated from the prompt “An image of Boudica.”

The example adds two observations to the Platypus case. First, because the summary says nothing about Boudica’s appearance, appearance propositions such as “The woman has blue markings on her face” can be neither confirmed nor refuted. The errors that the judge does find come from world knowledge, such as the Roman armour, or from the sex of the person. Second, the GPT Image 2 image contains the entity name, which the blind judge can read (Section 5.4.2).

4.6.3 A City: Belgrade

Belgrade is a Random entity from the Famous tier, so only RQ3 applies. Its lead summary states that Belgrade is the capital of Serbia, that it lies at the confluence of the Sava and the Danube, and that about 1.2 million people live in the city proper. Figure 4.6 shows the three images. GPT Image 2 draws a riverside old town at sunset with moored river boats and a tall baroque church tower; the view closely resembles Belgrade’s old town seen from the Sava, with the bell tower of St. Michael’s Cathedral. FLUX.2-dev draws a Central European river city with a stone arch bridge, and FLUX.2-klein-4B an aerial view of a city with green copper domes, red roofs, and a clock tower. Neither of the two FLUX images shows a landmark that identifies Belgrade.

(a) GPT Image 2

(b) FLUX.2-dev

(c) FLUX.2-klein-4B

Figure 4.6: Images generated from the prompt “An image of Belgrade.” All three receive zero false propositions.

The judge sees no difference between them. All three blind descriptions consist of similar propositions, for example “A tall church spire is visible in the skyline” (GPT Image 2), “A stone

Table 4.5: Complete judge output for the Boudica images in Figure 4.5, in the format of Table 4.4. For FLUX.2-dev, the explanation names two of the four propositions counted as false; the judge does not say which other two it counted.

Image and decisions	Propositions (blind call) and explanation (grounded call)
GPT Image 2 type: Person RQ1 ✓ RQ2 ✓ false: 0/8	(1) The central figure is a woman with long red hair. (2) The woman has blue markings on her face. (3) The woman is holding a spear. (4) The woman is wearing a plaid cloak. (5) The woman is standing in a carved wooden chariot. (6) There is a red banner with the text 'BOUDICA' and a triskelion symbol. (7) There are several men in armor in the background. (8) The sky is cloudy. *Explanation:* none.
FLUX.2-dev type: Person RQ1 ✓ RQ2 ✗ false: 4/9	(1) A woman is wearing a metal helmet with a plume. ✗ (2) The woman is holding a long spear in her right hand. (3) The woman is holding a round shield in her left hand. (4) The woman is wearing a metal breastplate and greaves. ✗ (5) The woman is wearing a red cape. (6) There are multiple soldiers in similar armor in the background. (7) There are red flags visible behind the soldiers. (8) The setting is a rocky, mountainous terrain. (9) The sky is overcast with a muted color palette. *Explanation:* "The woman is depicted in Roman-style armor (helmet, breastplate, greaves), which contradicts her role as a leader of an uprising against the Roman Empire."
FLUX.2-klein-4B type: Person RQ1 ✗ RQ2 ✗ false: 2/9	(1) A man is standing in a mountainous region. ✗ (2) The man is wearing a brown fur-lined garment. (3) The man is wearing a crown on his head. (4) The man is holding a spear in his right hand. (5) The man is holding a spear with a feather in his left hand. (6) There is a body of water in the background. (7) There are green mountains in the background. (8) The man has a beard. ✗ (9) The text 'BUULDCTH' is overlaid at the bottom of the image. *Explanation:* "The image depicts a man with a beard, whereas Boudica was a queen."

arch bridge crosses the river" (FLUX.2-dev), and "There are multiple buildings with green domes" (FLUX.2-klein-4B). The grounded call finds no false proposition in any of them: 0 of 7, 0 of 8, and 0 of 7. Nothing in the summary contradicts a river, a bridge, or a dome, and nothing in it confirms the church tower that makes the GPT Image 2 image recognizable. For RQ3 the three images are therefore equally good, although only one of them shows Belgrade. Section 5.3 returns to this case.

4.6.4 Recurring Failure Patterns

Bad generations follow four recurring patterns, listed in Table 4.6. Three of the six examples, including Dennis Dart (Figure 1.1), are Random entities without an annotated type or feature, so the table compares all examples only by their false propositions.

The first pattern is the most severe, because the model draws a different kind of thing altogether. On the common set, every proposition is judged false for 54 of 864 images (6.3%; the three Montezuma images are excluded, see Section 5.4.2): 13 for GPT Image 2, 12 for FLUX.2-dev, and 29 for FLUX.2-klein-4B. In 24 of these 54 images the model drew a person. The last pattern is the hardest to see in the numbers, because an image that asserts almost nothing about the entity cannot contradict the lead summary. The FLUX.2-dev image for *Freedom of religion in Austria* passes every check, yet it shows nothing about religious freedom. For such images a hallucination rate of zero means "nothing false", not "knowledge shown".

Table 4.6: Recurring patterns of bad generations, with examples from the corpus. The last column gives false propositions out of all propositions for the image.

Pattern	Entity (model)	What the image shows	False
Wrong referent: the name is read as something else	Dennis Dart (FLUX.2-klein-4B)	a man, “Dennis” read as a first name	5/5
	Echidna (GPT Image 2)	an anime-style woman at a tea table, most likely the anime character of the same name	8/8
	Pretzel (GPT Image 2)	a dachshund with a name tag	8/8
Near miss: a related entity of the same type	Platypus (FLUX.2-klein-4B)	a capybara	5/5
Wrong detail: right entity, false attribute	Heath Streak (GPT Image 2)	a cricketer in a shirt with a New Zealand logo; Streak played for Zimbabwe	1/7
Generic image: nothing checkable	Freedom of religion in Austria (FLUX.2-dev)	an Alpine valley with the title written across it	0/5

5 Discussion

This chapter interprets the results. Section 5.1 answers the research questions, and Section 5.2 compares the findings with what is known about factual knowledge in language models. Section 5.3 critically assesses the methodology, explains why the extrapolation does not separate the models, and outlines a better experiment design. Section 5.4 discusses implications and limitations.

5.1 Answers to the Research Questions

RQ1. The three models draw the correct coarse type for 73–79% of the 97 curated entities of the common set, and the differences between them are not significant. Type recognition is close to perfect for flora and fauna (92–100%) and lowest for cultural artifacts and foods (32–40%); the latter mainly reflects the generic label "Object" assigned in the blind call, as the much higher feature rates for this category show (Section 4.2).

RQ2. The models differ strongly in whether they show what distinguishes an entity from similar ones: GPT Image 2 does so for 62.9% of the curated entities, FLUX.2-dev for 46.4%, and FLUX.2-klein-4B for 28.9%. The gap between RQ1 and RQ2 is largest for the smallest model, which recognizes the type of 73.2% of the entities but shows the distinctive feature for fewer than a third. For these models, knowing the class of an entity is much easier than knowing how it differs from its neighbours, as the Platypus example shows (Figure 4.4).

RQ3. Of the propositions extracted from the images, 91.0% (GPT Image 2), 90.3% (FLUX.2-dev), and 82.3% (FLUX.2-klein-4B) are not contradicted by the lead summary. This share is an upper bound on the share of true propositions, because the judge cannot confirm what the lead summary does not mention. For GPT Image 2 the share of false propositions grows from 3.0% for Famous to 23.0% for Long-tail entities, although its Long-tail images contain fewer propositions; nine of its 13 images in which every proposition is false are Long-tail entities. Its errors on rare entities thus come from drawing the wrong thing, not from describing more. The FLUX models show no consistent popularity effect.

5.2 Comparison with Knowledge in Language Models

The population-weighted extrapolation in Table 4.3 puts the number of non-contradicted visual propositions over all English Wikipedia articles at about 43 million for each model. This is the same order of magnitude as the 101 million triples that GPTKB extracts from GPT-4o-mini (Hu et al., 2025b), but the two numbers measure different things. GPTKB covers 2.9 million entities with about 35 triples each; the extrapolation assumes one image with about six non-contradicted propositions for each of 7.2 million articles. About a third of these propositions describe the scene or transcribe text rather than the entity (Section 3.4), so the entity-related part is closer to 25–30 million. Finally, a GPTKB triple is a relational fact elicited by many prompts, whereas a proposition is only what one image happens to show. Moreover, the extrapolated number hardly depends on the model, although the models differ clearly in RQ2; Section 5.3 shows that it reflects the design of the measure more than the knowledge of the generators.

The per-model results nevertheless point in the direction known from language models. The three generators recognize the coarse type about equally often (73–79%, no significant difference), but differ strongly on distinctive features (63%, 46%, and 29%). The RQ2 ranking follows model size: GPT Image 2, whose size is undisclosed but whose predecessor was built into GPT-4o, ranks first; FLUX.2-dev, with a 24B language model and a 32B image transformer, ranks second; FLUX.2-klein-4B, with 4B parameters in each component, ranks last. With three models this is an observation and not a scaling result, because size, training data, and architecture change together, and the two FLUX models differ in both components at once. It is consistent with the view that coarse types are available even to small models, while entity-specific visual detail requires more capacity. The popularity effect for GPT Image 2 (Table 4.2) likewise resembles the drop in LLM accuracy for rare entities (Mallen et al., 2023; Kandpal et al., 2023).

5.3 Critical Assessment of the Methodology

Section 5.1 reports clear differences between the models: GPT Image 2 shows the distinctive feature for 62.9% of the curated entities and FLUX.2-klein-4B for 28.9%. The extrapolation in Section 4.5, by contrast, credits all three models with about 43 million non-contradicted propositions. If both numbers measured the same knowledge, they could not both be right. This section argues that the contradiction comes from the design of the extrapolation measure, not from the models, and outlines an experiment design that would avoid it.

5.3.1 Why the Extrapolation Does Not Separate the Models

The extrapolation multiplies, for each tier, the number of propositions per image by the share of them that is not contradicted (Equation 3.2). Neither factor depends much on whether the model knows the entity.

The number of propositions is set by the judge. The blind call asks for "a list of short, atomic factual claims" about the image, and the judge answers with lists of similar length for almost every image: 7.4 propositions on average with a standard deviation of 1.4, and between 7.1 and 7.7 per model (Table 4.1). A recognizable image of the requested entity and

an image of something else receive lists of about the same length. On the curated subset (without Montezuma), images that show the distinctive feature yield 6.8 propositions and images that do not yield 7.0. The count therefore describes the answering style of the judge more than the knowledge of the generator.

Not contradicted is not the same as known. A proposition counts towards the total unless the lead summary shows it to be false. Most propositions concern details that the summary does not mention, so they pass whether the image shows the right entity or not. On the 96 curated entities of the common set (without Montezuma, Section 5.4.2), an image that fails RQ2 still contributes 6.19 non-contradicted propositions, compared with 6.73 for an image that passes. The Belgrade images (Figure 4.6) show the same effect: a recognizable view of Belgrade and two generic cities all receive zero false propositions. For historical figures, whose summaries describe what a person did rather than how the person looked, 58 of the 66 images in the common set contain no false proposition at all. Only gross errors, such as a man for a queen or a capybara for a platypus, reduce the count noticeably.

The weighting magnifies the least informative tier. The population weights assign 94% of Wikipedia to the Long-tail tier, which therefore contributes 93–94% of each total. In this tier, two effects happen to cancel out: GPT Image 2 has the highest error rate (23.0%) but also the longest descriptions, so all three models end up at about 5.9 non-contradicted propositions per image (Section 4.5). The tiers in which GPT Image 2 is clearly ahead carry too little weight to matter. The near equality of the totals thus rests on 65 Long-tail entities per model and on a coincidence of two opposite effects.

These points do not invalidate the answers to RQ1–RQ3. RQ2 asks whether one pre-specified, entity-specific fact is shown, and this question separates the models. The extrapolation asks how many statements about an image survive a check that can only refute, and this quantity is nearly constant. A simple variant on the curated subset makes the difference visible. If non-contradicted propositions are counted only for images that show the distinctive feature, the average per image falls from 6.66, 6.45, and 6.21 to 4.32, 2.97, and 1.90 for GPT Image 2, FLUX.2-dev, and FLUX.2-klein-4B. The ratio between the best and the worst model grows from 1.07 to 2.3, close to the ratio of their RQ2 rates. The totals in Table 4.3 should therefore be read as the number of visual statements that the models can make without being refuted by a short reference text, not as a measure of how much they know.

Two further weaknesses concern the grounded call itself and are visible in the examples of Section 4.6. The call returns only a count and a free-text explanation, so it is not recorded which propositions were counted as false (Table 4.5); and the judge sometimes rejects all propositions of an image once it decides that the image shows a different entity (Table 4.4). The RQ2 decision, in addition, depends on how the blind call happens to word its description.

5.3.2 A Better Experiment Design

The main weakness lies in the direction of the check. The current design starts from the image: the judge describes whatever it sees, and the reference can only refute. A design that measures knowledge should start from the entity: it should fix in advance what a model that knows the entity would draw and then check how much of it the image shows. The following changes would implement this, and most of them reuse parts of the existing pipeline.

1. **Reference-derived checklists.** For each entity, a fixed list of visual facts would be collected before generation: from the full Wikipedia article, from Wikidata properties such as colour, material, architectural style, or occupation, and from reference photographs, in the spirit of TIFA (Hu et al., 2023) and T2I-FactualBench (Huang et al., 2025c). The judge would answer one question per fact, for example "Does the animal have a duck-like bill?". The score is the share of checklist facts that the image shows. It has the same denominator for every model, it cannot be raised by longer descriptions, and RQ2 is the special case of a checklist with a single item. The number of facts shown per entity could then be extrapolated in the same way as in Section 3.7.
2. **Three labels per proposition.** Where open-ended descriptions are kept, the grounded call should label each proposition separately as *supported*, *contradicted*, or *not verifiable*, and only supported propositions would count as knowledge. The current design merges the first and the third label. A reference image, as in T2I-FactualBench, would make appearance verifiable; there, removing it lowered the agreement with human ratings from 0.57 to 0.44.
3. **A generic baseline.** Propositions such as "There is a river" or "The man has a beard" are true of many entities. Generating one image from the type alone ("An image of a city", "An image of a queen") and running the same pipeline would show how many propositions a model produces without any entity-specific knowledge; only the excess over this baseline would count. Alternatively, the judge could be asked whether a proposition distinguishes the entity from other entities of the same type.
4. **Identification instead of description.** A blind judge could be shown the image together with the entity name and several distractors of the same type, for example five European capitals for Belgrade, and asked which of them the image shows. Identification accuracy above chance would measure entity-specific knowledge directly and would not depend on how many propositions the judge writes. Text written into the image would have to be masked first, since it often names the entity (Section 5.4.2).
5. **Sampling and validation.** Because the Long-tail tier carries 94% of the weight, an extrapolation needs many more Long-tail entities than the 65 used here, or a sample drawn in proportion to the population. Several images per entity would capture the variation between generations, and a sample of propositions annotated by humans would show how often the judge is right, as T2I-FactualBench did with three annotators.

Within the scope of this thesis, the existing pipeline answers RQ1, RQ2, and RQ3 as they are defined, and the experiments were not repeated with the design above. The extrapolation, however, would need such a reference-first design before it could be compared with knowledge counts for language models such as GPTKB.

5.4 Implications and Limitations

5.4.1 Implications

Visual plausibility is insufficient for using T2I outputs in knowledge-oriented settings. A generated image may get the entity type right while failing to include the feature that makes the entity recognizable. Conversely, an image can look detailed while containing false contextual details. Proposition-level checking is therefore a useful complement to image-quality metrics.

The examples in Section 4.6 also show that the benchmark score depends on how directly an entity can be drawn. For a concrete entity such as Dennis Dart, a wrong image produces several explicit contradictions: a hand-held device or a portrait of a man cannot be a bus chassis. For an abstract entity such as Freedom of religion in Austria, models draw landscapes or captioned scenes that contradict nothing in the summary. The hallucination rate should therefore be read as a measure of false visual statements, not as a measure of how much the image shows about the entity.

5.4.2 Limitations

What RQ3 measures. The judge can refute a proposition but not confirm it: propositions that the lead summary does not mention are not counted, so the complement of the hallucination rate is only an upper bound on the share of true propositions (Section 5.3.1).

Judge. All decisions are made by one open-weight model, `gemma-4-31B-it`, rather than a large proprietary evaluator such as GPT-4V (Zhang et al., 2023), and its decisions were not validated against human judgements. The judge is also not fully consistent: it accepted the label "Object" as a type match for artifacts and foods in 27 of 64 cases and rejected it in the others. Since all three generators were scored by the same judge, the ranking of the models is expected to be more robust than the absolute percentages.

Reference. The Wikipedia lead summary is short, so a visible detail can be true without being mentioned. Title resolution also picked the wrong article for two curated entities: "Montezuma" resolved to a small community in New Mexico and "Ada Lovelace" to an Nvidia GPU architecture. For Montezuma, all three models drew an Aztec ruler with a feathered headdress, but the judge marked the type as wrong and counted 8, 8, and 9 hallucinations. Correcting this entity would raise each model's RQ1 rate by about one percentage point and lower each overall hallucination rate by about 0.4 points; the ranking of the models does not change. The Random entity "Stardust" is probably affected in the same way.

Blind call. The blind call does not see the prompt, but it does see any text written inside the image. Models often write the entity name into the picture: this happens in about 47% of the GPT Image 2 images, 24% of the FLUX.2-dev images, and 14% of the FLUX.2-klein-4B images (keyword count). In these cases the judge can read the name, which makes RQ1 easier than intended and favours GPT Image 2 most.

Sample. The curated subset is small (97 entities in the common set) and was annotated by one person. Each entity was generated once per model without a fixed seed, so the results do not capture the variation between repeated generations. The Random subset is stratified by design and does not reproduce the distribution of Wikipedia; Section 3.7 corrects for this only in the extrapolation.

Refusals. GPT Image 2 refused nine prompts, including three historical figures. The common set keeps these refusals from being scored as errors, but conclusions about GPT Image 2 apply only to the entities it generated, and the missing entities are not a random sample.

6 Conclusion and Future Work

This thesis introduced a benchmark for the factual knowledge of text-to-image models and applied it to GPT Image 2, FLUX.2-dev, and FLUX.2-klein-4B. Three findings stand out. First, recognizing the type of an entity is not the same as knowing it: the models draw the correct type for 73–79% of the curated entities, but the distinctive feature for only 29–63%. Second, 9–18% of the visual propositions contradict the lead summary, and for GPT Image 2 this share grows from 3% for popular to 23% for rare entities. Third, the most severe errors come from reading the entity name as something else, such as a man called Dennis instead of a bus model; the image then looks convincing and is entirely wrong. Compared with language models, whose factual knowledge has been materialized at the scale of 10^8 triples, the knowledge of image generators is harder to elicit and limited to what can be drawn.

The extrapolation to the English Wikipedia credits all three models with about 43 million non-contradicted propositions, although they differ clearly in RQ2. The critical assessment in Section 5.3 traces this to the design of the measure: the judge writes about seven propositions for almost any image, and a short reference text can refute only a few of them, so the count hardly depends on whether the model knows the entity. The extrapolated number is therefore an upper bound on unrefuted visual statements, not an estimate of visual knowledge.

Future work should reverse the direction of the check: it should derive a checklist of visual facts for each entity from reference texts, knowledge-base properties, and reference images before generation, and measure how many of them an image shows. Labelling each proposition as supported, contradicted, or not verifiable, subtracting a baseline generated from the entity type alone, and asking a blind judge to identify the entity among distractors would separate entity knowledge from generic content. The judge should be validated against human annotations on a sample of propositions, and more than one judge should be used. Repeated generations per entity, a closed label set for the blind call, more models, and a larger, naturally sampled Long-tail set would allow stronger conclusions.

A Supplementary Evaluation Information

A.1 Missing Entity–Model Pairs

Table A.1 lists the missing evaluations identified from the generation failure log. All rows belong to GPT Image 2 and all are safety refusals. The complete CSV register, including provider error details, is stored with the analysis outputs.

Table A.1: Missing GPT Image 2 evaluations.

Entity	Recorded reason
Mahatma Gandhi	Safety refusal
Frida Kahlo	Safety refusal
Salvador Dalí	Safety refusal
Transformers: Age of Extinction	Safety refusal
Joker: Folie à Deux	Safety refusal
Hugh Greenwood	Safety refusal
Holby City episode 1102	Safety refusal
Treehouse of Horror	Safety refusal
Edouard Wojtczak	Safety refusal

A.2 Reproducibility Files

The benchmark entities are stored in the entity CSV. Generated images are organized by model and category. The evaluation results contain one row per successful entity–model pair, and the full benchmark release includes the analysis script, the evaluation outputs, and the generated image folders. The analysis script regenerates coverage tables, common-set comparisons, tier-level hallucination tables, proposition-count tables, correlations, the extrapolation diagnostics used in Section 5.3, and figures from these inputs. This separation between raw rows and derived tables allows the reported values to be recomputed after changing a metric or adding another model.

The public code and generated assets are available at `https://github.com/mrEkso/t2i-factuality-benchmark`.

A.3 Judge Prompts

Both calls use `gemma-4-31B-it` with temperature 0, JSON output, and the system message "You are a strict JSON-only API. Always return valid JSON without markdown wrapping." Placeholders are shown in angle brackets.

Call 1 (blind). The image is attached to the message.

```
You are an objective computer-vision classifier. You are NOT told what
this image is supposed to depict - judge only what is visually present.

Output a raw JSON object with exactly these keys:
1. "predicted_type": The primary object type shown (e.g., Building,
   Person, Animal, Landscape, Object, None).
2. "description": A brief text description of the main visual elements.
3. "propositions": A list of short, atomic factual claims about what is
   visually shown (e.g., ["The building has a round tower", "There is a
   river in the foreground"]).
```

Call 2 (grounded, curated entities). No image is attached. For Random entities, the expected type and feature and keys 1 and 2 are omitted.

```
You are a strict fact-checking judge. You are given:
- The real-world entity name: "<entity>"
- Reference text about this entity (from its Wikipedia summary):
  """<Wikipedia lead summary>"""
- An expected object type for this entity: "<expected type>"
- An expected unique feature for this entity: "<expected feature>"
- Visual observations made by a separate, blind image classifier (it did
  NOT know the entity name), about a generated image that is supposed to
  depict this entity:
  predicted_type = "<predicted type>"
  description = "<description>"
  propositions = [<propositions>]

Using ONLY the ground truth text above (plus general knowledge only if
the ground truth text is marked unavailable), output a raw JSON object
with exactly these keys:

1. "rq1_type_match": Boolean. Does predicted_type match the expected
   reality of "<entity>"?
2. "rq2_feature_present": Boolean or null. Do the propositions/description
   clearly indicate the feature "<expected feature>"? If expected_feature
   is "N/A", output null.
3. "rq3_hallucinations_count": Integer. How many propositions are FALSE
   or contradicted by the ground truth text?
4. "rq3_hallucinations_details": Short string explaining which
   propositions are wrong, or "None" if all are consistent with the
   ground truth.
```

A.4 Curated Entities and Annotations

Tables A.2 to A.5 list the 100 curated entities with the expected type and the distinctive feature used for RQ1 and RQ2 (Section 3.1). The annotations are given exactly as they appear in the entity file.

Table A.2: Curated entities: Famous Landmarks and Architecture.

Entity	Expected type	Distinctive feature
Neuschwanstein	Castle	Romanesque Revival architecture on a rugged hill
Parthenon	Temple	Doric marble columns
Chichen Itza	Pyramid	Step pyramid with central stairs
Machu Picchu	Citadel	Dry-stone walls on a mountain ridge
Stonehenge	Monument	Circular ring of standing stones
Colosseum	Amphitheatre	Elliptical shape with ruined outer wall
Petra	Archaeological site	Rock-cut temple facade
Acropolis	Citadel	Ancient ruins on a rocky outcrop
Alhambra	Palace complex	Moorish architecture with intricate tile work
Borobudur	Temple	Multiple stupas shaped like perforated bells
Pantheon	Temple	Massive concrete dome with a central oculus
Louvre	Museum	Glass pyramid structure in the courtyard
Atomium	Building	Nine interconnected steel spheres
Guggenheim	Museum	Curving titanium-clad exterior
Alcatraz	Prison	Fortified structures on an isolated island
Versailles	Palace	Extensive geometric gardens and classical facade
Monticello	Plantation house	Neoclassical dome and red brick exterior
Potala	Palace	Massive white and red fortress-like walls on a hill
Karnak	Temple complex	Massive sandstone pillars with hieroglyphics
Teotihuacan	Archaeological site	Avenue of the Dead aligned with massive pyramids
Ephesus	Ancient city	Library of Celsus facade with Roman columns
Persepolis	Ancient capital	Monumental staircases and bull reliefs
Knossos	Palace	Red Minoan columns and bull-leaping frescoes
Pompeii	Ancient city	Volcanic ash ruins and preserved Roman streets
Sigiriya	Fortress	Massive column of rock with ruined foundations on top

Table A.3: Curated entities: Historical Figures.

Entity	Expected type	Distinctive feature
Rasputin	Mystic	Wild unkempt beard and dark robes
Cleopatra	Egyptian Queen	Distinctive braided hair and heavy dark eyeliner
Saladin	Sultan	Medieval Muslim turban and chainmail armor
Ada Lovelace	Mathematician	Victorian dress and hair in braided side buns
Montezuma	Emperor	Elaborate feathered headdress
Mahatma Gandhi	Civil rights leader	Bald head with round wire-rimmed glasses and a dhoti
Vercingetorix	Chieftain	Long Gallic mustache and ancient Celtic armor
Bismarck	Chancellor	Pickelhaube helmet and walrus mustache
Nelson Mandela	President	Grey hair and colorful patterned Madiba shirt
Hatshepsut	Pharaoh	Female pharaoh with a false beard
Boudica	Queen	Long red hair and ancient tribal clothing
Frida Kahlo	Painter	Distinctive unibrow and traditional Mexican dress
Che Guevara	Revolutionary	Black beret with a single red star
Dante	Poet	Red robe and a laurel wreath
Paganini	Violinist	Tall thin figure with wild hair playing a violin
Tokugawa Ieyasu	Shogun	Traditional samurai armor and chonmage topknot
Ludwig van Beethoven	Composer	Messy grey hair and red scarf
Sitting Bull	Chief	Single feather in braided hair and bone choker
Vincent van Gogh	Painter	Red hair and bandaged ear
Salvador Dali	Painter	Thin upward-pointing mustache
Nefertiti	Queen	Flat-topped blue crown
Edgar Allan Poe	Writer	Dark messy hair and a thin mustache
Leo Tolstoy	Writer	Long flowing white beard and peasant blouse
Nikola Tesla	Inventor	Dark hair parted in the middle and a thin mustache
Galileo Galilei	Astronomer	Beard and a Renaissance-era telescope

Table A.4: Curated entities: Rare Flora and Fauna.

Entity	Expected type	Distinctive feature
Axolotl	Amphibian	External feathery gills on the side of its head
Shoebill	Bird	Massive bulbous bill resembling a shoe
Rafflesia	Flower	Massive five-petaled speckled red fleshy petals
Okapi	Mammal	Zebra-like stripes on legs with a dark brown body
Pangolin	Mammal	Body covered in overlapping keratin scales
Narwhal	Marine mammal	Long spiral tusk protruding from its head
Quokka	Mammal	Small furry body with a prominent smiling facial expression
Olm	Amphibian	Blind snake-like pale body with external gills
Platypus	Mammal	Duck-like bill and beaver-like tail
Tarsier	Primate	Enormous eyes that are larger than its brain
Saiga	Mammal	Over-sized flexible tubular snout
Aye-aye	Primate	Unusually long and thin skeletal middle finger
Kakapo	Bird	Flightless green plumage and owl-like facial disk
Welwitschia	Plant	Two continuously growing long torn leaves on desert ground
Baobab	Tree	Extremely thick cylindrical trunk with sparse top branches
Coelacanth	Fish	Lobed fins and armor-like deep blue scales
Tuatara	Reptile	Spiny crest down its back and lizard-like appearance
Wombat	Mammal	Stout barrel-shaped body and short legs
Echidna	Mammal	Spiky coat with a slender beak-like snout
Cassowary	Bird	Large bony casque on top of a bright blue neck
Capybara	Mammal	Large boxy muzzle and reddish-brown fur
Ocelot	Mammal	Chain-like rosettes and spots on golden fur
Fossa	Mammal	Slender cat-like body with a very long tail
Binturong	Mammal	Shaggy black hair and long white whiskers
Quetzal	Bird	Vibrant green plumage and an extremely long tail

Table A.5: Curated entities: Specific Cultural Artifacts and Foods.

Entity	Expected type	Distinctive feature
Beshbarmak	Food	Boiled flat noodles with minced meat
Haggis	Food	Savory pudding encased in a round casing
Matryoshka	Toy	Nested painted wooden female figures
Astrolabe	Instrument	Brass circular ancient astronomical mechanism
Macaron	Confection	Two smooth meringue halves with a ruffled base
Pretzel	Pastry	Symmetrical twisted knot shape with salt grains
Croissant	Pastry	Crescent-shaped flaky golden layers
Paella	Food	Yellow saffron rice cooked in a wide shallow pan
Kimchi	Food	Red spicy fermented cabbage leaves
Moai	Statue	Oversized monolithic stone head with prominent brow
Antikythera	Mechanism	Corroded ancient bronze gear wheels
Ushabti	Figurine	Small ancient Egyptian funerary mummy-like figure
Kokeshi	Toy	Wooden figure with a simple trunk and no arms
Boomerang	Weapon	Curved aerodynamic wooden shape
Ramen	Food	Noodles in dark broth topped with sliced pork
Baklava	Pastry	Layers of flaky phyllo dough filled with chopped nuts
Poutine	Food	French fries topped with cheese curds and brown gravy
Eclair	Pastry	Oblong choux dough filled with cream and topped with chocolate
Escargot	Food	Cooked snails served in their spiral shells
Churros	Pastry	Star-shaped fried dough sticks coated in sugar
Sombrero	Hat	Wide brim and a high pointed crown
Katana	Sword	Curved single-edged blade with a long grip
Didgeridoo	Instrument	Long wooden cylindrical tube painted with dot art
Kilt	Clothing	Knee-length pleated skirt with a tartan pattern
Kimono	Clothing	T-shaped wrapped garment with wide square sleeves

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